

Interim Report — Week 3 Challenge

End-to-End Insurance Risk Analytics & Predictive Modeling

10 Academy: Artificial Intelligence Mastery

Author: Lalise Fufi

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1. Executive Summary

This project focuses on analyzing historical South African auto insurance data from AlphaCare Insurance Solutions (ACIS) to uncover risk patterns, profitability trends, and opportunities for data-driven insurance pricing optimization.

The project combines:

- Exploratory Data Analysis (EDA)
- Data quality assessment
- Insurance risk analytics
- Data Version Control (DVC)
- Reproducible machine learning pipelines

Key findings from the interim analysis include:

- Significant profitability differences across provinces
- High claim concentration in Gauteng, KwaZulu-Natal, and Western Cape
- Weak correlation between premiums and claims
- Presence of major financial outliers in claims-related variables
- Increasing claims trend over time
- Successful implementation of a reproducible DVC pipeline

The project establishes a strong foundation for upcoming hypothesis testing and predictive risk modeling tasks.

2. Introduction

Insurance companies rely heavily on historical claims data to:

- Evaluate customer risk
- Optimize premium pricing
- Improve profitability

- Reduce underwriting uncertainty

AlphaCare Insurance Solutions (ACIS) aims to improve its marketing and pricing strategy using evidence-driven analytics instead of intuition-based decisions.

This project analyzes 18 months of insurance data (February 2014 – August 2015) to identify:

- Low-risk customer segments
- High-risk geographic regions
- Vehicle categories associated with elevated claims
- Patterns affecting insurance profitability

The project also emphasizes reproducibility through Data Version Control (DVC), which is essential in financial and insurance analytics environments.

3. Business Problem and Objectives

Business Problem

ACIS is preparing for expansion within the South African auto insurance market. However, inefficient pricing and limited understanding of risk patterns can lead to:

- Reduced profitability
- Poor customer targeting
- Underpriced high-risk customers
- Loss-making insurance portfolios

Traditional pricing approaches may fail to capture:

- Geographic risk variation
 - Vehicle-related claim behavior
 - Customer demographic risk patterns
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Project Objectives

The project aims to:

- Analyze insurance claims and premium data
- Identify high-risk and low-risk customer segments
- Evaluate profitability using Loss Ratio and Margin
- Build reproducible analytics pipelines using DVC
- Prepare data for predictive modeling and risk-based pricing

To achieve this, the project:

- Performed Exploratory Data Analysis (EDA)
 - Conducted data quality assessment
 - Built reusable data processing utilities
 - Configured GitHub Actions CI/CD
 - Implemented Data Version Control (DVC)
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4. Dataset Overview

The dataset contains:

- Policy information
- Customer demographics
- Vehicle characteristics
- Geographic data
- Insurance cover details
- Premium and claims records

Time Period

- February 2014 – August 2015
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Key Metrics

Loss Ratio

$\text{Loss Ratio} = \text{TotalClaims} / \text{TotalPremium}$

Measures insurance portfolio profitability.

Margin

$\text{Margin} = \text{TotalPremium} - \text{TotalClaims}$

Measures policy-level contribution profit.

5. Exploratory Data Analysis (EDA)

5.1 Data Summarization

EDA was conducted to understand:

- Data distributions

- Risk behavior
- Profitability trends
- Geographic variations

The analysis included:

- Descriptive statistics
- Data type inspection
- Numerical feature exploration

Key variables analyzed:

- TotalPremium
- TotalClaims
- SumInsured
- CustomValueEstimate

5.2 Data Quality Assessment

The dataset was evaluated for:

- Missing values
- Duplicate records
- Incorrect data types
- Data inconsistencies

Handling Strategy

- Numerical missing values handled using median imputation
- Duplicate records removed
- Date columns converted into datetime format
- Data cleaning automated using reusable Python scripts

Data Cleaning Automation

The cleaning process was implemented in:

src/data_loader.py

Pipeline execution:

```
python src/data_loader.py
```

5.3 Univariate Analysis

Performed:

- Histograms

- Distribution analysis
- Categorical frequency analysis

Key Findings

- TotalClaims is heavily right-skewed
 - SumInsured contains extreme outliers
 - A few provinces dominate the insurance portfolio
 - Claims-related variables exhibit large variance
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5.4 Bivariate and Multivariate Analysis

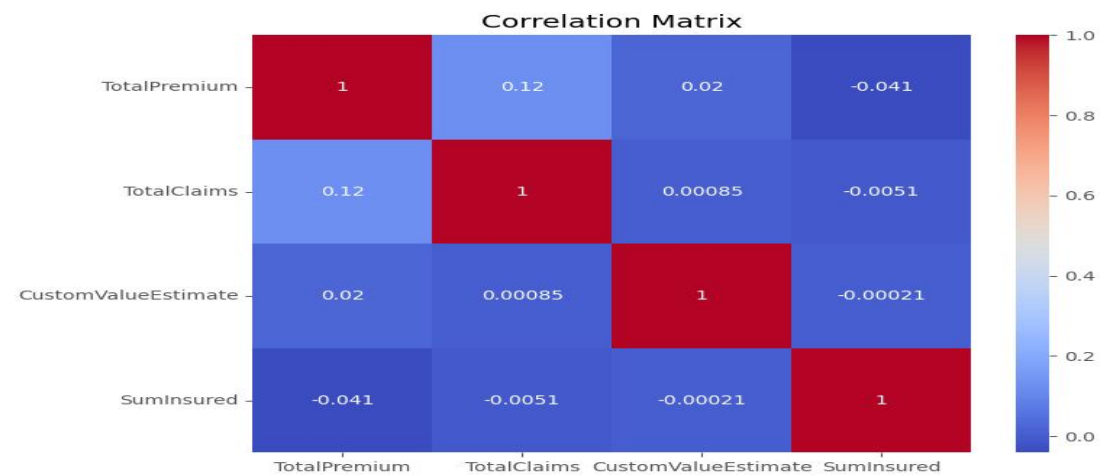
Performed:

- Correlation analysis
- Scatter plots
- Province-level premium vs claims comparisons

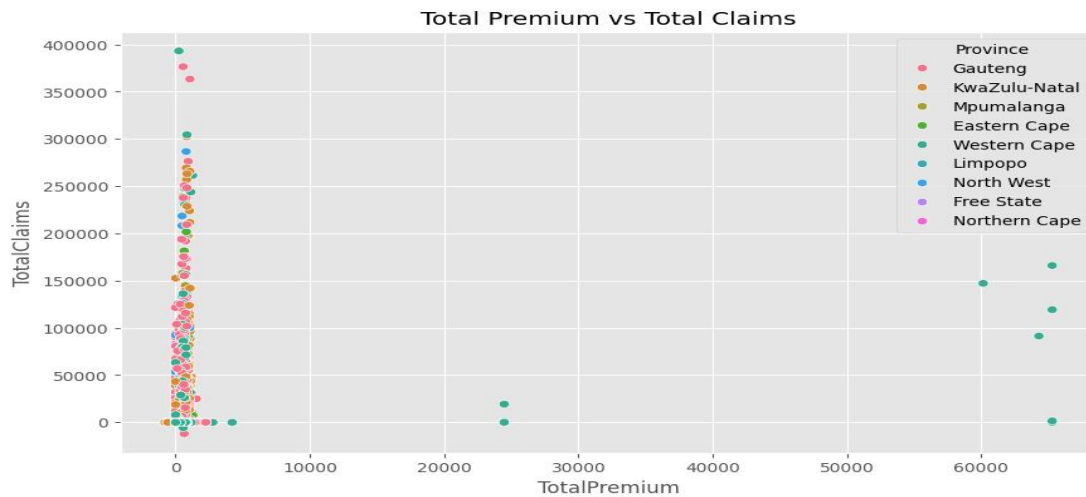
Key Findings

- Weak correlation exists between TotalPremium and TotalClaims
 - Geographic regions show major profitability differences
 - Some provinces consistently generate higher claims relative to premiums
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Visual Output — Correlation Heatmap



Visual Output — Premium vs Claims Scatter Plot



5.5 Geographic Trends

The analysis compared:

- Claims by province
- Premium distribution
- Vehicle types across regions

Key Findings

High-risk provinces:

- Gauteng
- KwaZulu-Natal
- Western Cape

More profitable provinces:

- Northern Cape
- Eastern Cape
- Limpopo

5.6 Outlier Detection

Outlier analysis was conducted using:

- Boxplots
- Distribution visualizations

Major Outliers Detected

- TotalClaims

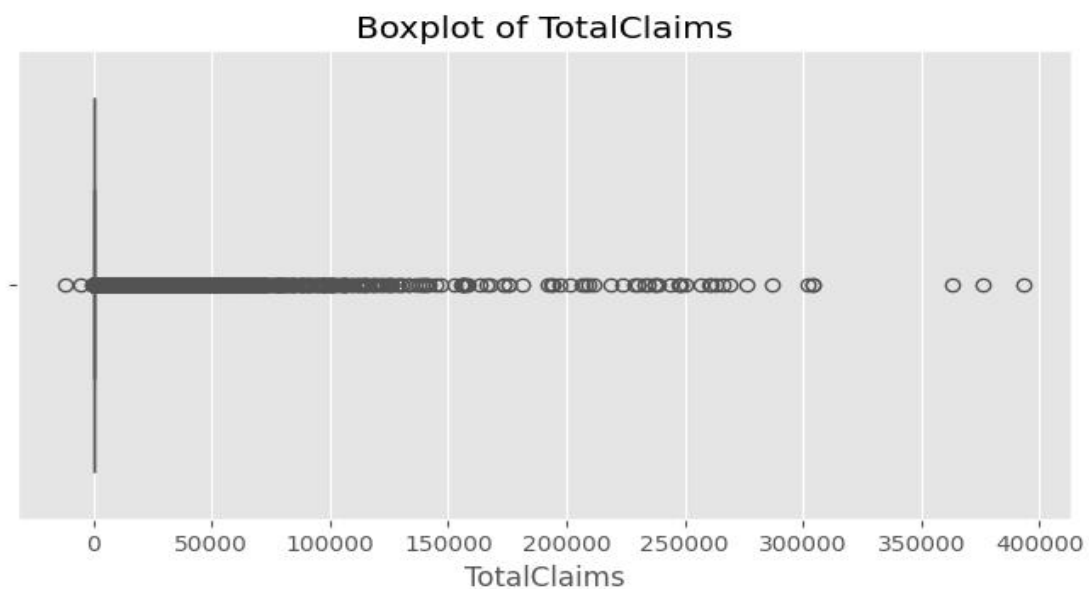
- SumInsured
- CustomValueEstimate

Business Implications

These outliers may indicate:

- Fraudulent claims
- Severe accidents
- High-value vehicles
- Data inconsistencies

Visual Output — Outlier Boxplots



6. Key Business Questions Answered

6.1 What is the overall Loss Ratio?

Loss Ratio was used as the primary profitability indicator.

Findings

- Some provinces exhibited loss ratios greater than 1
- Gauteng showed the highest loss ratio
- Northern Cape showed the strongest profitability

6.2 How does Loss Ratio vary?

By Province

Highest risk:

- Gauteng
- KwaZulu-Natal
- Western Cape

Lowest risk:

- Northern Cape
 - Limpopo
 - Eastern Cape
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By Gender

- “Not specified” gender category showed the highest loss ratio
 - Female customers generally showed lower loss ratios
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By Vehicle Type

Certain vehicle categories generated disproportionately higher claims.

6.3 Are there significant outliers?

Yes.

Major outliers were found in:

- TotalClaims
- SumInsured
- CustomValueEstimate

These may strongly influence:

- Correlation analysis
 - Mean calculations
 - Predictive modeling performance
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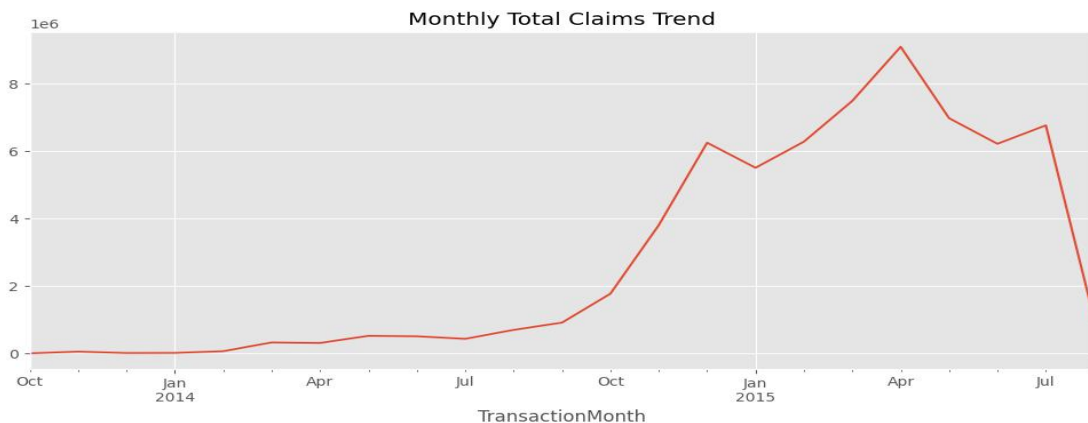
6.4 Are there temporal trends?

Yes.

Claims analysis over time showed:

- Increasing claims frequency
- Accelerating claim severity
- Potential structural risk changes after late 2015

Visual Output — Claims Trend Over Time



6.5 Which vehicle makes generated the highest claims?

Vehicle makes associated with higher claims included:

- SUZUKI
- JMC
- HYUNDAI

Insight

SUZUKI appeared as a major outlier with significantly elevated average claim values.

7. Data Version Control (DVC)

7.1 Objective

DVC was implemented to create:

- Reproducible pipelines
- Version-controlled datasets
- Auditable workflows
- Reliable collaboration processes

This is especially important in:

- Insurance analytics
 - Financial systems
 - Machine learning environments
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7.2 DVC Workflow

Initialize DVC

```
dvc init
```

Configure Local Remote Storage

```
dvc remote add -d localstorage C:\dvc-storage
```

Track Dataset

```
dvc add data/MachineLearningRating_v3.txt
```

Reproduce Pipeline

```
dvc repro
```

Push Data to Remote Storage

```
dvc push
```

Pull Data

```
dvc pull
```

7.3 Reproducible Pipeline

A reproducible pipeline was created using:

```
stages:
  clean_data:
    cmd: python src/data_loader.py
    deps:
      - src/data_loader.py
      - data/MachineLearningRating_v3.txt
    outs:
      - data/insurance_data_cleaned.csv
```

The pipeline automatically:

- Detects dependency changes
- Re-runs affected stages
- Updates outputs

- Maintains reproducibility
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8. GitHub Actions CI/CD

A CI/CD workflow was configured using GitHub Actions.

Pipeline Includes

- Dependency installation
- Flake8 linting
- Automated testing

Triggered on:

- Push events
 - Pull requests
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CI Issues Encountered and Resolved

Several CI issues were encountered during implementation:

- Trailing whitespace failures
- CRLF vs LF line-ending conflicts
- GitHub Actions authentication issues
- Flake8 formatting errors

These issues were resolved through:

- Line ending normalization
 - File cleanup
 - Workflow correction
 - Improved repository configuration
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9. Key Insights

Insight 1 — Geographic Profitability Differences

High-premium provinces were not always profitable.

Example

- Gauteng generated high premiums but also very high claims.

Insight 2 — Weak Premium-Claims Relationship

Premiums did not strongly correlate with claims, suggesting:

- Potential pricing inefficiencies
- Need for improved risk segmentation

Insight 3 — Significant Claims Growth

Claims increased substantially over time.

Possible causes:

- Increased portfolio exposure
- Rising claim severity
- Risk concentration

Insight 4 — Outlier Influence

Extreme claims may heavily distort:

- Statistical analysis
- Model training
- Profitability metrics

Robust preprocessing will be necessary in later stages.

10. Interim Deliverables Completed

Task 1 — EDA

Completed:

- Data summarization
- Data quality assessment
- Univariate analysis
- Multivariate analysis
- Geographic trend analysis
- Outlier detection
- Insight-driven visualizations

Task 2 — DVC

Completed:

- DVC installation and initialization
- Local remote storage setup
- Dataset tracking
- Reproducible pipeline creation
- DVC reproduction testing
- Documentation updates

11. Challenges Encountered

Several implementation challenges were encountered during Task 2:

- DVC tracking conflicts with existing outputs
- GitHub Actions CI failures due to whitespace and newline formatting
- File line-ending inconsistencies between Windows and Linux environments
- Dataset parsing issues caused by incorrect separators

These issues were resolved through:

- Pipeline restructuring
- Proper DVC output management
- Line ending normalization
- Improved preprocessing scripts

12. Future Work

Task 3 — Hypothesis Testing

Upcoming work includes:

- Statistical validation of risk differences
- Province-level risk analysis
- Gender and zip-code comparisons

Task 4 — Predictive Modeling

Upcoming modeling tasks include:

- Claim severity prediction
 - Risk-based pricing models
 - Random Forest and XGBoost modeling
 - SHAP interpretability analysis
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13. Conclusion

This interim project successfully established the foundation for end-to-end insurance risk analytics and predictive modeling.

The EDA revealed:

- Significant geographic risk variation
- Profitability imbalances
- High-impact financial outliers
- Weak premium-claims relationships

In addition, a reproducible DVC pipeline and CI/CD workflow were successfully implemented, ensuring scalable and auditable analytics workflows.

The project now provides a strong baseline for statistical testing and machine learning-driven insurance pricing optimization.

14. References

- DVC Documentation
- Scikit-learn Documentation
- SHAP Documentation
- GitHub Actions Documentation
- Insurance Analytics Resources
- 10 Academy Week 3 Challenge Materials